

Catfishes and relatives

PHYLUM	Chordata
CLASS	Osteichthyes
SUBCLASS	Actinopterygii
SUPERORDER	Ostariophysi
ORDERS	5
FAMILIES	85
SPECIES	About 10,500

Catfishes and their relatives account for three-quarters of all species of freshwater fishes. Although they are a varied group, almost all species have a bony internal structure, known as the Weberian apparatus, which is used in sound reception and contributes to their acute hearing. They are divided into 5 groups: milkfishes and relatives, cyprinids (the most widespread

of which are carp and minnows), characins (including tetras, tigerfishes, and piranhas), neotropical knifefish, and catfishes themselves. Although mostly confined to freshwater, some species are found in brackish water or the sea.

Anatomy

Catfishes and their relatives vary greatly in size and shape. Most are small fishes, less than 4 in (10 cm) long, although some species reach 16 ft (5 m). The Weberian apparatus that unites most members of this group consists of modified vertebrae that act as a set of levers to transmit sound waves received by the swim bladder to the inner ear. This feature gives catfishes and their relatives an acute sense of hearing, and many species use sound to communicate with each other.

Catfishes typically have a flattened head with a wide mouth that is usually surrounded by

whiskerlike sensory barbels; these are used for touch and taste, and they compensate for poor eyesight. Most species have smooth, scaleless skin but

in several groups the scales have developed into a dense armor. Nearly all species have sharp or serrated pectoral and dorsal spines, some of which can release venom.

Characins and cyprinids look similar to one another, although most characins are small and often brightly colored, while cyprinids are generally dull-colored and can reach lengths of up to 6½ ft (2 m). Another distinguishing factor is the arrangement of the teeth: characins have teeth in their jaws, often with a replacement row directly behind the functional teeth, whereas cyprinids have toothless jaws but usually have teeth in the pharynx. Characins, unlike cyprinids, also have a small fin (called an adipose fin) between the dorsal fin and tail, which is not supported by rays and stores fat.

Milkfishes and their relatives (which include the beaked salmon) are the only members

of this group that do not have a Weberian apparatus. Instead, vibrations are transmitted from the swim bladder to the inner ear by a modified set of ribs.

Feeding

This group includes voracious predators, scavengers, herbivores, and filter feeders. Catfishes are generally predators or scavengers, living close to the bottom in murky water, although a few species are herbivores. Most characins are schooling fishes that eat plants or prey on invertebrates or fishes: piranhas are notorious meat-eaters. Most species of cyprinids have downward-turned, often suckerlike mouths, with one or more pairs of sensory barbels on either side. They usually feed on detritus or on small invertebrates and plants in the sediment; few are predatory. Milkfishes are filter feeders but some of their relatives feed on small animals.

NIGHT HUNTER

Most catfishes are active at night and feed alone. The barbels around the mouth are covered with taste buds, which help them find food. This long-whiskered (or pimelodid) catfish is found in tropical rivers in South America. The larger members of this group feed on fishes and other animals, including monkeys that fall into the water.

